

DMAIC Control Plan — US Flight Delay Reduction

Project: Flight Delay Root Cause Analysis

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CTQ METRIC

- Metric: On-Time Arrival Rate (%)
- Definition: Flight arrives ≤ 15 min of scheduled time
- Baseline: 79.94% | DPMO: 200,600 | Sigma: 2.3
- Target: 87.00% | DPMO: 130,000 | Sigma: 2.7

MEASUREMENT SYSTEM

- Data Source: BTS On-Time Performance Database (transtats.bts.gov)
- Frequency: Monthly
- Tool: Power BI Dashboard (automated KPI monitoring)

CONTROL METHODS

1. Monitor monthly on-time rate via Power BI SPC chart
2. Trigger investigation if any month falls below LCL (66.12%)
3. Flag airlines with on-time rate $< 75\%$ for carrier review
4. Escalate if summer months (Jun-Aug) drop $> 5\%$ below baseline

REACTION PLAN

- If rate drops below LCL: Convene delay root cause review within 5 days
- If specific airline drops $> 10\%$: Issue carrier performance warning
- If weather events spike: Activate pre-cancellation protocol

SUSTAINABILITY NOTE

- Each 1% improvement in on-time rate = ~67,600 fewer delayed flights/year
- Estimated fuel savings: ~\$12M/year (avg 11 gal wasted fuel per delay hour)
- CO2 reduction: ~115 metric tons/year per 1% improvement